

Hospital Readmission Risk Predictor:

A Phased MLOps Build

Exploratory Data Analysis Report

Prepared for: Hospital Administration & Care Management Stakeholders

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<https://github.com/AngelineSetiawan/AngelineSetiawan-hospital-readmission-risk-predictor>

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1. Background & Question

Hospitals across the United States are financially penalized under the Centers for Medicare & Medicaid Services' (CMS) Hospital Readmissions Reduction Program (HRRP) when their 30-day readmission rates exceed expected benchmarks. Roughly one in fourteen eligible hospitals loses at least one percent of Medicare reimbursement under this program, with penalties capped at three percent of payments (Centers for Medicare & Medicaid Services, n.d.). For a hospital administration or care-management team, this translates into a direct, recurring financial exposure that is only partially within the team's control once a patient has already been discharged. The value of a readmission-risk model is that it moves the point of intervention earlier: if a care team can identify which patients are most likely to bounce back within 30 days before those patients ever leave the building, they can target discharge planning, follow-up calls, and medication review at exactly the patients who need it most.

This project's research question is whether a patient's risk of 30-day readmission can be predicted from routinely captured clinical encounter data, and whether that prediction can be delivered through a live, self-updating system rather than a one-time report. This second half of the question matters as much as the first: a model that scores well in a notebook but is never re-evaluated against new data quietly goes stale, and a stale risk score is arguably worse than no score at all, because it creates false confidence. This project therefore treats the predictive model as one component of a larger operational pipeline that logs every experiment, promotes only the best-performing model into production, serves predictions through a live API, and, as a stretch goal, retrains and redeploys itself on a schedule.

The working hypothesis is that patients with more complex or intensive care histories — longer

hospital stays, more prior inpatient or emergency visits, and a higher number of medications — are more likely to be readmitted within 30 days, because these factors reflect greater underlying disease severity and more complicated post-discharge needs. The corresponding prediction is that a supervised classification model will outperform a naive majority-class baseline on standard imbalanced-classification metrics (AUC-ROC and recall for the readmitted class), and that utilization-related features — number of prior inpatient visits, time in hospital, and number of medications — will rank among the model's most important predictors. This exploratory data analysis is a first, direct check on that hypothesis: before any model is trained, does the raw data show these utilization variables moving in the expected direction for patients who were, in fact, readmitted within 30 days? No changes to the research question or hypothesis were made following proposal feedback; the analysis below is the first empirical test of the reasoning laid out in that proposal.

2. Methods

2.1 Dataset

The analysis draws on the Diabetes 130-US Hospitals for Years 1999–2008 dataset, hosted by the UCI Machine Learning Repository (Clore et al., 2014). The dataset spans ten years of clinical care across 130 U.S. hospitals and integrated delivery networks, and it contains 101,766 inpatient diabetic encounters with more than fifty fields per encounter, including demographics, admission and discharge codes, diagnosis codes, medication counts, lab and procedure counts, and utilization in the year prior to each encounter. It is licensed CC BY 4.0 and was obtained as a direct CSV download from the UCI repository (`diabetic_data.csv`), so no API access, credentialing, or web scraping was required.

¹A second, companion file supplied by UCI, `IDS_mapping.csv`, is not encounter-level data at all — it is a small reference table that stacks three separate ID-to-label lookup tables (one each for admission type, discharge disposition, and admission source) with a blank line marking the boundary between them. This file was parsed programmatically into three lookup tables and joined onto the main encounter table on the corresponding coded ID columns, producing three new, human-readable label columns (`admission_type_id_label`, `discharge_disposition_id_label`, `admission_source_id_label`) while leaving the original numeric codes untouched for later modeling use. This join is the only data-preparation step involving more than one source file, and it is what makes the categorical summaries in Section 3 legible to a non-technical audience rather than a table of bare integer codes.

All acquisition and preparation code — including the `IDS_mapping` join logic, the data-source attributions and citations, and inline comments explaining each step — is version-controlled in the project's GitHub repository (see title page), together with the data dictionary reproduced in the Appendix (Section A). That repository is structured so another analyst could reproduce every table and figure in this report directly from the original UCI download, rather than from an ad hoc or undocumented notebook.

This dataset was selected over more recently collected alternatives after a deliberate review of what is realistically available. Modern hospital EHR extracts with real readmission outcomes are protected health information under HIPAA and are not released publicly; the closest research-grade modern alternative, MIMIC-IV, requires completed human-subjects research training, a signed data use agreement, and PhysioNet credentialing that could not be secured within this project's timeframe. Synthetic modern alternatives avoid that gatekeeping but do not

carry real readmission labels, which would undermine the project's predictive validity. Two limitations of this choice are worth stating plainly for planning purposes. First, the encounters were collected between 1999 and 2008 — before the HRRP existed and before widespread electronic health record adoption — so any relationships the eventual model learns describe historical patterns in this population and should be read as a proof of concept rather than a claim about current hospital performance. Second, after collapsing the outcome into a binary 30-day label, the positive (readmitted) class shrinks to roughly 11 percent of encounters, which is a meaningful but not enormous sample for model training and will require an evaluation approach that does not rely on raw accuracy.

2.2 EDA Methods

Consistent with the EDA stage of this project, no missing-value imputation and no row deletion were performed at this point. Three narrow, non-destructive steps were taken solely to make the data explorable: the dataset's placeholder "?" values were converted to proper null values so that missingness could be counted accurately; the native three-class readmission label (readmitted in under 30 days, readmitted after 30 days, or not readmitted) was collapsed into the binary target `readmitted_30d` used throughout this report; and one variable, `weight`, was dropped because it is missing for roughly 97 percent of encounters and is not usable in any form at this sample size. No other column was altered, dropped, or imputed.

Descriptive statistics were generated for the eight core continuous predictors most relevant to the hypothesis (length of stay, lab procedures, non-lab procedures, medications, and three prior-utilization counts, plus number of diagnoses) and for a set of key categorical fields spanning demographics, admission and discharge context, test results, and medication activity. For continuous variables this included measures of central tendency and spread (mean, median,

standard deviation, minimum, and maximum) as well as skew and excess kurtosis to flag heavily one-sided or long-tailed distributions before they reach a model. For categorical variables this included frequency counts and percentages for every observed category. Visual methods included histograms to inspect distribution shape, boxplots and a rate-by-bucket line chart to compare hypothesis-relevant predictors directly against the readmission outcome, a bar chart of readmission rate by age bracket to check whether age behaves as expected, and a Pearson correlation heatmap (correlogram) across the continuous predictors and the target to screen for multicollinearity before any feature-selection or regularization decisions are made. A Spearman rank-correlation supplement was also computed against the target to catch monotonic-but-nonlinear relationships that a purely linear Pearson correlation could understate — this matters here because several utilization counts are heavily zero-inflated.

3. Results

The dataset's target variable is highly imbalanced: as Figure 1 shows, only 11.2% of the 101,766 encounters (11,357 encounters) resulted in readmission within 30 days, while the remaining 88.8% did not. This imbalance is the single most important fact shaping the analysis plan going forward — it is the reason evaluation metrics later in this project will lean on AUC-ROC and recall for the positive class rather than raw accuracy, since a model that always predicted "not readmitted" would already be 88.8% accurate while being clinically useless.

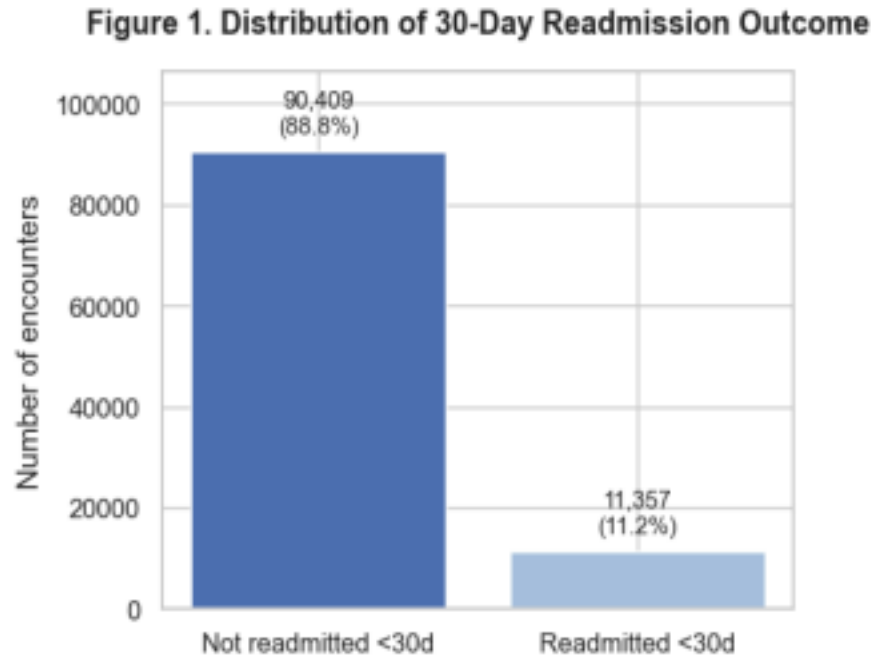


Figure 1. Distribution of the 30-day readmission outcome across all 101,766 encounters.

Only about one in nine encounters ends in a 30-day readmission. This class imbalance means accuracy alone will be a misleading metric for the eventual classifier, and it is the primary justification for prioritizing AUC-ROC and recall in later model evaluation. Note: this figure is computed on the full, uncleaned sample and still includes the 1,652 encounters ending in patient death (Section 4.2); because those encounters are guaranteed non-readmissions, the true positive rate is expected to shift slightly upward once they are excluded during preprocessing.

3.1 Descriptive Statistics

Table 1 summarizes the eight continuous predictors most relevant to the stated hypothesis. Two patterns stand out. First, the three prior-utilization counts — `number_outpatient`, `number_emergency`, and `number_inpatient` — are extremely right-skewed (skew values of 8.83, 22.86, and 3.61, respectively) with very heavy tails (excess kurtosis in the hundreds to over a thousand for `number_emergency`). In practical terms, most patients have zero prior outpatient, emergency, or inpatient visits in the preceding year, while a small number of patients have dozens. This zero-inflation will need to be accounted for during feature engineering, either through capping, binning, or a model family that tolerates skewed inputs well. Second,

num_medications and time_in_hospital are also right-skewed but far less extreme, which is consistent with the fact that hospital stays and medication counts are bounded by clinical and administrative limits in a way that outpatient visit counts are not.

Variable	N	Mean	Median	Std. Dev.	Min	Max	Skew	Kurtosis
time_in_hospital	101,766	4.40	4.0	2.99	1	14	1.13	0.85
num_lab_procedures	101,766	43.10	44.0	19.67	1	132	-0.24	-0.25
num_procedures	101,766	1.34	1.0	1.71	0	6	1.32	0.86
num_medications	101,766	16.02	15.0	8.13	1	81	1.33	3.47
number_outpatient	101,766	0.37	0.0	1.27	0	42	8.83	147.90
number_emergency	101,766	0.20	0.0	0.93	0	76	22.86	1,191.63
number_inpatient	101,766	0.64	0.0	1.26	0	21	3.61	20.72
number_diagnoses	101,766	7.42	8.0	1.93	1	16	-0.88	-0.08

Table 1. Descriptive statistics for the eight core continuous predictors ($n = 101,766$).

Table 2 summarizes all twelve categorical variables carried into this EDA — every categorical field named in the analysis plan is represented, covering demographics, admission and discharge context, glucose/HbA1c testing, and diabetes medication activity. For each variable it reports how many categories it takes on, its single most common category, and the missingness or "unknown" rate that applies to it, whether that missingness comes from a true null or from a coded placeholder label such as "NULL" or "Not Available" (see Table 6 below). The full 90-row, category-by-category frequency breakdown that this summary is drawn from — needed to see every individual discharge-disposition or admission-source category, for example — is reproduced in full in the Appendix (Section B) and is also version-controlled in the GitHub repository. Read together, Table 2 shows a patient population that skews older (58.5% of encounters are age 60 or above), is admitted predominantly through emergency or urgent channels (71.2% combined) rather than as scheduled elective care, and is rarely lab-tested for glucose or HbA1c at this encounter (94.8% and 83.3% not tested, respectively) — a pattern more

likely to reflect testing protocol than patient health and worth carrying into the modeling stage as its own signal rather than discarding as "missing."

Variable	# Categories	Most Common Category	% Most Common	% Missing / Unknown
race	6	Caucasian	74.78%	2.23%
gender	3	Female	53.76%	0.00%
age (bracket)	10	[70–80)	25.62%	0.00%
admission_type_id_label	8	Emergency	53.05%	10.22%
discharge_disposition_id_label	26	Discharged to home	59.19%	4.60%
admission_source_id_label	17	Emergency Room	56.50%	6.66%
max_glu_serum	3 + not tested	Not tested	94.75%	0.00%
AlCresult	3 + not tested	Not tested	83.28%	0.00%
change (med. changed)	2	No change	53.80%	0.00%
diabetesMed (Rx present)	2	Yes	77.00%	0.00%
insulin	4	No	46.56%	0.00%
metformin	4	No	80.36%	0.00%

Table 2. Categorical variable summary — all twelve categorical fields in the analysis plan ($n = 101,766$). Full category-level frequencies are in the Appendix (Section B).

3.2 Hypothesis-Relevant Comparisons

Table 3 stratifies the core continuous predictors by readmission outcome and speaks most directly to the hypothesis. Patients readmitted within 30 days show a higher mean on every one of the eight utilization variables than patients who were not readmitted, and the gap is largest for number_inpatient (mean of 1.12 prior inpatient visits among the readmitted group versus 0.56 among the non-readmitted group — roughly double). Time in hospital, medication count, and emergency visits all move in the same direction, though the differences in means are more modest for those variables. This is an encouraging early signal for the hypothesis, though it should be read cautiously: these are simple two-group mean comparisons, not controlled effect sizes, and the medians for several variables (time_in_hospital, num_procedures) are identical across groups even where the means differ, which suggests the difference is being driven by the

distribution's tail rather than a shift in the typical patient.

Continuous predictor	Not readmitted (mean / median)	Readmitted <30d (mean / median)
time_in_hospital	4.35 / 4.0	4.77 / 4.0
num_lab_procedures	42.95 / 44.0	44.23 / 45.0
num_procedures	1.35 / 1.0	1.28 / 1.0
num_medications	15.91 / 15.0	17.01 / 16.0
number_outpatient	0.36 / 0.0	0.39 / 0.0
number_emergency	0.18 / 0.0	0.36 / 0.0
number_inpatient	0.56 / 0.0	1.12 / 0.0
number_diagnoses	7.35 / 8.0	7.99 / 8.0

Table 3. Continuous predictor means and medians, stratified by 30-day readmission outcome.

Table 4 complements Table 3 by stratifying the same readmission outcome across five of the most policy-relevant categorical predictors, so that both the continuous and categorical sides of the population are checked against the target. Demographic splits (gender, race) show only a narrow 1–2 percentage-point spread in readmission rate across categories, suggesting demographics alone are unlikely to be strong standalone predictors. Treatment-related variables show more separation: patients who were prescribed a diabetes medication were readmitted at 11.63% versus 9.60% for those who were not, and patients whose medication regimen was changed during the encounter were readmitted at 11.82% versus 10.59% for those whose regimen was unchanged. Both gaps point in the same direction as the utilization-based hypothesis in Section 1 — greater treatment intensity or a mid-course medication change tracks with higher readmission risk — and both are plausible additions to the feature set alongside the utilization counts already highlighted in Table 3.

Predictor	Category	30-Day Readmission Rate	N (encounters)
Gender	Female	11.25%	54,708
Gender	Male	11.06%	47,055
Race	Caucasian	11.29%	76,099
Race	African American	11.22%	19,210
Race	Hispanic	10.41%	2,037
Race	Asian	10.14%	641
Race	Other	9.63%	1,506
Diabetes medication prescribed	Yes	11.63%	78,363
Diabetes medication prescribed	No	9.60%	23,403
Medication changed at encounter	Changed	11.82%	47,011
Medication changed at encounter	Not changed	10.59%	54,755
Admission type	Emergency	11.52%	53,990
Admission type	Urgent	11.18%	18,480
Admission type	Elective	10.39%	18,869

Table 4. 30-day readmission rate stratified by key categorical predictors, computed directly from the encounter-level data (n = 101,766).

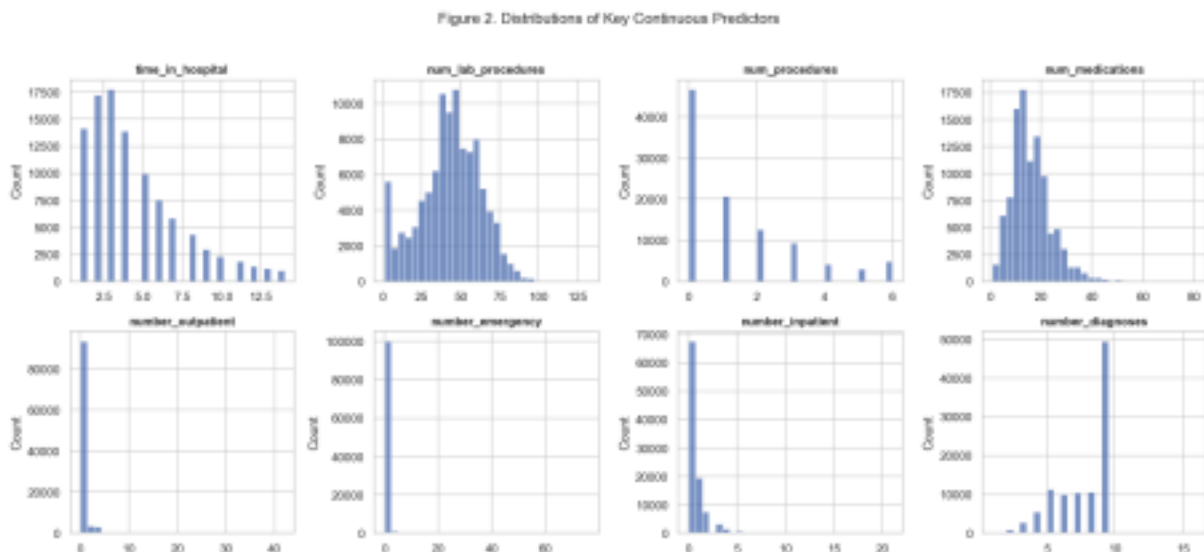


Figure 2. Distributions of the eight core continuous predictors.

Length of stay, lab procedures, and medication count are moderately right-skewed but roughly bell-shaped, while the three prior-utilization counts are dominated by a single spike at zero with a long, thin tail. Number_diagnoses is unusual in the opposite direction — most encounters cluster near the coding maximum of nine diagnoses, which likely reflects a data-entry convention worth flagging during cleaning rather than a true distribution of clinical complexity.

Figure 3. Hypothesis-Relevant Predictors by Readmission Outcome

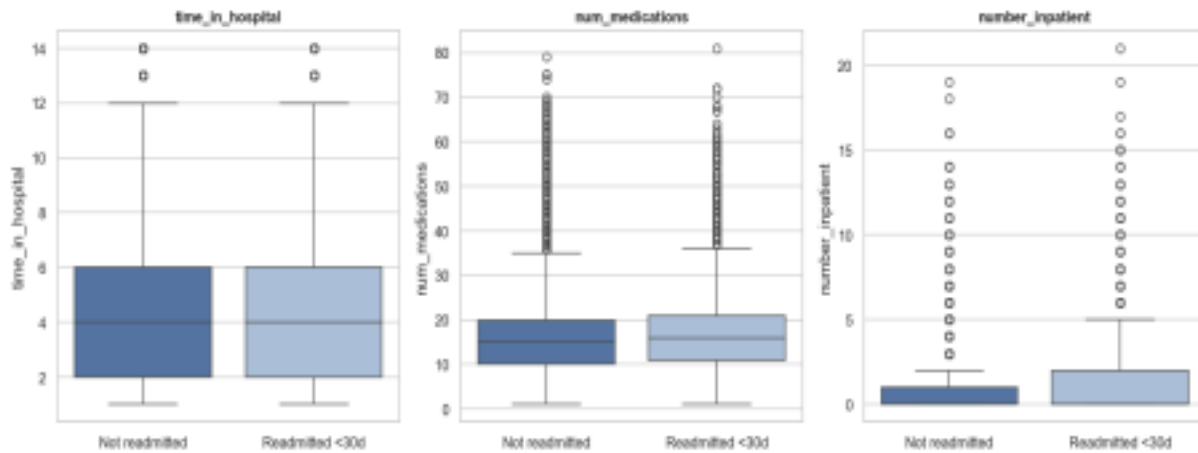


Figure 3. Time in hospital, medication count, and prior inpatient visits, compared by readmission outcome.

The boxplots make the same story visible without averaging it away: the median and interquartile range for `number_inpatient` shift up noticeably for readmitted patients, while `time_in_hospital` and `num_medications` shift only slightly. Prior inpatient utilization currently looks like the strongest of these three hypothesis-relevant predictors. Note: like Figure 1, this is computed prior to excluding the 1,652 structurally-negative expired encounters described in Section 4.2.

Figure 4. 30-Day Readmission Rate by Age Bracket

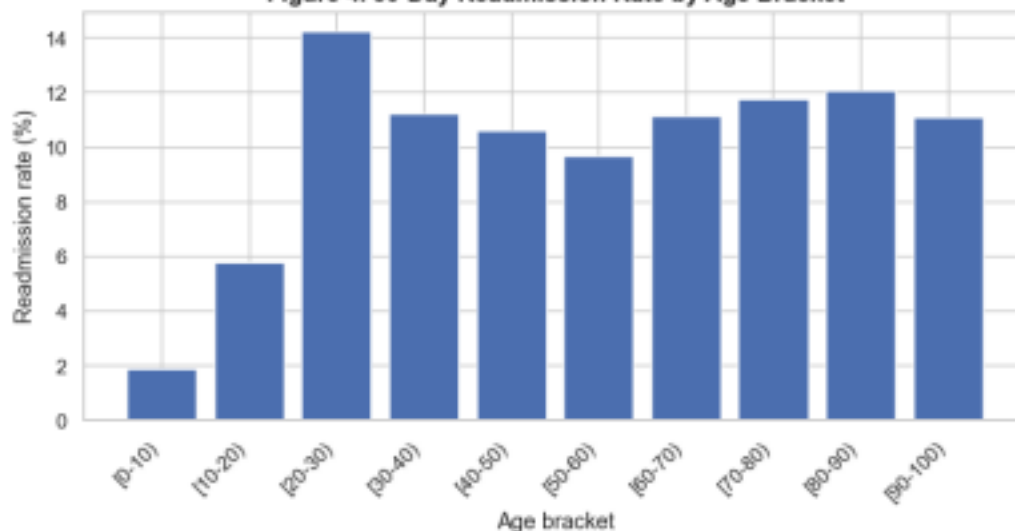


Figure 4. 30-day readmission rate by ten-year age bracket.

Readmission rate rises from under 2% in the youngest bracket to roughly 10–12% across most adult brackets, broadly consistent with age as a proxy for accumulating health burden. The one exception is the [20–30) bracket, where the rate spikes to just over 14%; this bracket is also one of the smallest in the dataset (1,657 encounters, 1.6% of the sample), so the estimate is noisier than the larger adult brackets and is flagged here for a closer look once the data is split for modeling, rather than treated as a confirmed pattern. Note: rates shown here are computed prior

to excluding the 1,652 structurally-negative expired encounters described in Section 4.2, so true age-bracket rates are expected to shift slightly upward, most for the older brackets where in-hospital deaths concentrate.

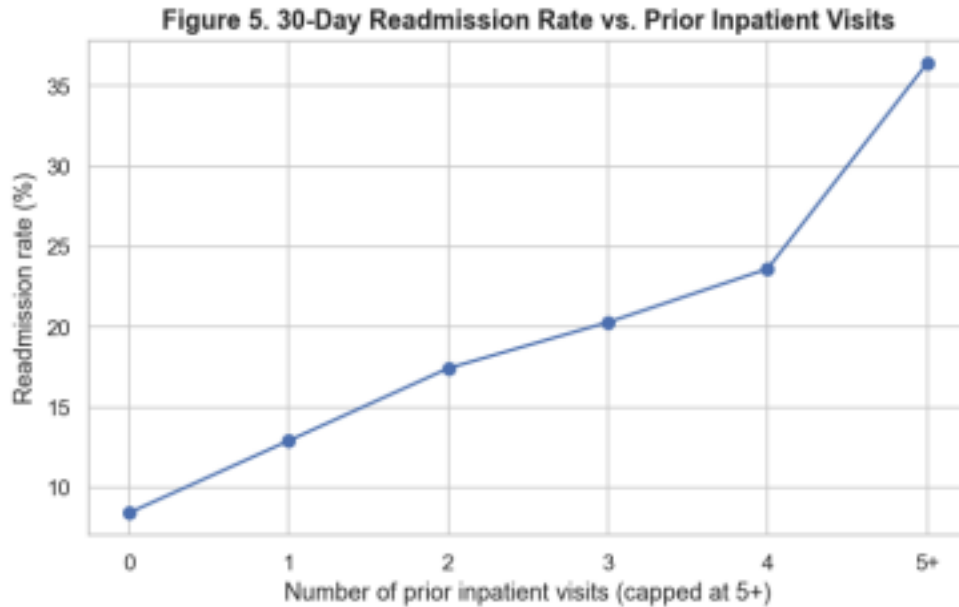


Figure 5. 30-day readmission rate versus number of prior inpatient visits (capped at 5+).

This is the clearest single piece of evidence for the hypothesis in the entire analysis: readmission rate climbs in an almost straight line from about 8% for patients with no prior inpatient visits to over 36% for patients with five or more, a more than four-fold increase. Prior inpatient utilization looks likely to be one of the most important predictors in the eventual model. Note: as in Figures 1, 3, and 4, these rates are computed prior to excluding the 1,652 structurally-negative expired encounters described in Section 4.2.

3.3 Missingness Encoding, as Implemented

The current "Preliminary handling plan" column in Table 5 states an intent (e.g., "encode as tested vs. not tested") without specifying the mechanics. This has since been implemented in the Phase 2 preprocessing notebook (Step 6), which additionally reasons through each field's likely missingness mechanism (MCAR, MAR, or MNAR) before choosing a treatment. The replacement column below states exactly what the notebook creates for each field.

Column	Missingness type	Encoding plan
max_glu_serum, A1Cresult	True null	Two engineered features per field: a binary <code>_tested</code> flag (1 = test was performed, 0 = not), and the categorical result itself, with "not tested" retained as its own explicit category rather than removed. Table 2 shows this is a testing-protocol signal, not noise, so it is preserved rather than imputed.
medical_specialty, payer_code	True null (39.6% / 49.1%)	One-hot or frequency <code>encode</code> the observed categories, plus an explicit "Unknown" category. Deletion is not viable given the missingness share; imputing a specific specialty or payer that was never recorded would fabricate information.
race	True null (2.2%)	Explicit "Unknown" category alongside the observed race categories. No separate indicator flag is needed at this low a missingness share.
diag_1, diag_2, diag_3	True null (<1.5% each)	Grouped into clinically meaningful ICD-9 category buckets per the existing plan, with an explicit "Unknown/Missing" bucket added to that same grouping scheme rather than handled separately.
admission_type_id_label, discharge_disposition_id_label, admission_source_id_label	Hidden missingness: coded strings ("NULL", "Not Available", "Not Mapped")	Collapse all coded-unknown variants into a single explicit "Unknown" category per field before one-hot or target encoding, so the model does not treat "Not Mapped" and "Not Available" as distinct, meaningful categories when both indicate an absence of information.

Table 5. Missingness inventory — Table 5. Missingness encoding plan by field ($n = 101,766$).

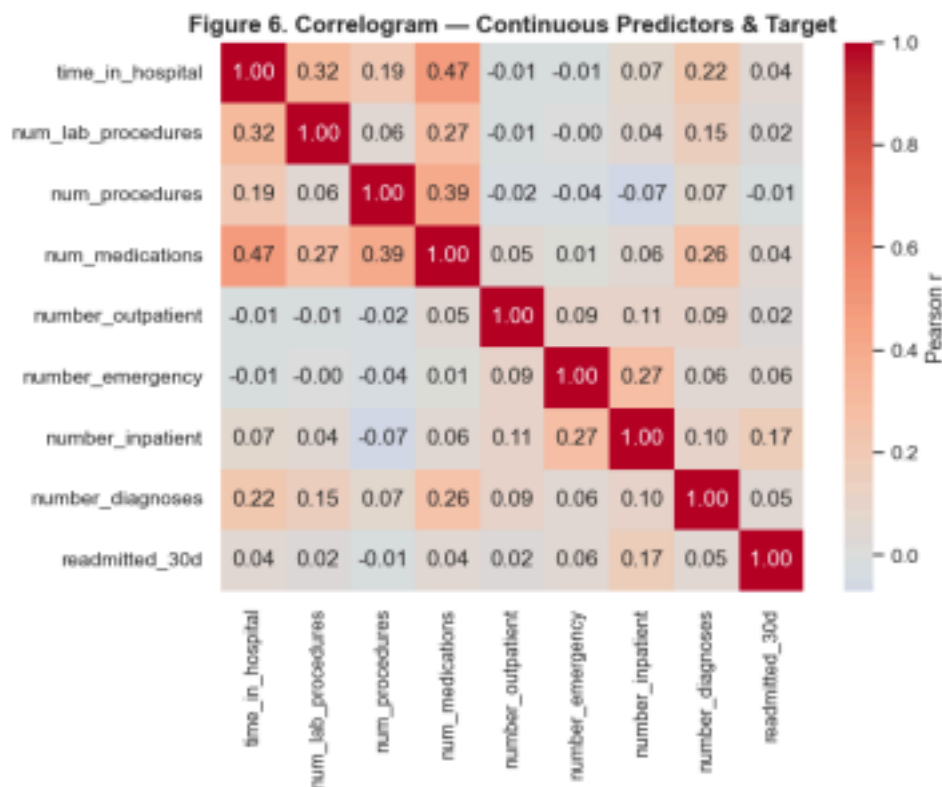
General principle to state alongside this table: no field is imputed with a mean, median, or mode. For every field listed above, the missingness itself appears informative (a testing-protocol choice, a coding gap, or a genuinely unrecorded value) rather than random, so it is preserved as its own explicit category or flag instead of being smoothed over.

Coded label column	Count coded "unknown"	% of encounters
admission_type_id_label	10,396	10.22%
discharge_disposition_id_label	4,680	4.60%
admission_source_id_label	6,781	6.66%

Table 6. Hidden missingness — coded "unknown" categories in the three ID-mapping label columns ($n = 101,766$).

3.4 Collinearity Assessment

Figure 6 shows the Pearson correlation matrix across all eight continuous predictors and the target, which satisfies the project's required collinearity check. Correlations among the predictors themselves are modest overall, with the strongest being `time_in_hospital` and `num_medications` ($r = 0.47$) and `num_procedures` and `num_medications` ($r = 0.39$) — both plausible, since longer or more procedure-heavy stays tend to involve more medications. None of the pairwise correlations reach a level (commonly cited thresholds are in the 0.7–0.8 range) that would raise serious multicollinearity concerns for a linear baseline model, though `time_in_hospital`, `num_lab_procedures`, `num_procedures`, and `num_medications` do form a loosely correlated cluster worth monitoring during feature selection.



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Figure 6. Correlogram of Pearson correlations across continuous predictors and the target (`readmitted_30d`).

Correlations with the target itself are uniformly weak in linear (Pearson) terms — the strongest is `number_inpatient` at $r = 0.17$ — which at first looks discouraging for the hypothesis. However, a Spearman rank-correlation check against the target tells a related but distinct story: `number_inpatient` still leads at 0.138, followed by `number_emergency` (0.063) and

time_in_hospital (0.051), preserving the same relative ordering. Neither correlation coefficient alone captures the step-function relationship visible in Figure 5, because *number_inpatient* is heavily zero-inflated; this is a reminder that weak linear correlation does not rule out a strong, non-linear relationship, and that tree-based models (random forest, XGBoost) are likely to extract more signal from these utilization variables than a linear correlation coefficient suggests.

3.5 Repeated-Measures Dependence Across Encounters

Every summary statistic and comparison reported in Sections 3.1 through 3.4 treats each of the 101,766 rows as an independent observation. That assumption does not hold for this dataset and is addressed directly here, before any modeling decisions are finalized.

The dataset's 101,766 encounters come from only 71,518 unique patients (*patient_nbr*). Table 3.5a summarizes the extent of repetition and its consequences for the target variable and for the utilization predictors central to the project's hypothesis.

Metric	Value	Why it matters
Patients with more than one encounter	16,773 (23.5% of all patients)	Nearly a quarter of the patient population is not independently represented
Share of all rows contributed by repeat patients	46.2% of 101,766 rows	Close to half the dataset consists of correlated, within-patient observations rather than independent encounters
30-day readmission rate, single-encounter patients	3.96%	The baseline rate for a patient seen only once is far below the reported overall figure
30-day readmission rate, repeat-encounter patients	19.55%	Roughly five times higher than single-encounter patients; this gap, not a uniform 11.2% rate, is the real structure in the data
Correlation: patient's total encounter count vs. <i>number_inpatient</i>	$r = 0.64$	The strongest predictor identified in Section 3 is mechanically tied to how many times a patient appears in the dataset at all
Share of positive-class rows from patients with >1 positive encounter	35.8%	The effective number of independent readmission events is smaller than the row count of 11,357 positive cases suggests

Table 3.5a. Summary of repeated-encounter structure in the dataset ($n = 101,766$ encounters, 71,518 unique patients).

Interpretation

Two consequences follow directly from these figures, and both revise how Sections 3.2 and 3.4 of this report should be read.

- Overstated independence in the target variable. The overall 11.2% readmission rate reported in Figure 1 is an average across two structurally different subpopulations: a majority of patients (76.5%) who are seen once and readmitted at under 4%, and a minority (23.5%) who are seen repeatedly and readmitted at nearly 20%. A model, or a simple hypothesis test, that treats every row as an independent draw from a single 11.2% base rate will understate the true uncertainty in any estimated relationship and can overstate how confidently a pattern generalizes to a new patient.
- A confounded predictor. The $r = 0.64$ correlation between a patient's total encounter count and `number_inpatient` means the strongest utilization predictor identified in Table 1 is, in part, measuring how many times a patient shows up in this dataset rather than an independent clinical signal at each visit. This does not invalidate the hypothesis, but it means the current EDA cannot yet distinguish "prior inpatient visits predict this specific readmission" from "this is a chronic high-utilizer, and chronic high-utilizers are both more likely to be readmitted and more likely to have a high `number_inpatient` count by construction."

This distinction matters operationally as well as statistically. A model that has learned to flag chronic high-utilizers is still clinically useful, but it is a different and narrower claim than the one stated in the project's hypothesis, and the two should not be conflated when the results are eventually presented to hospital administration.

Why the planned patient-grouped split does not resolve this

The time-aware, patient-grouped validation strategy described in Section 4.2 groups splits by `patient_nbr` so that no single patient's encounters appear in both the training and test partitions. *"Patient grouping in the split prevents information leakage between partitions; it does not, by itself, correct for correlation among repeated observations within a partition, which is why the primary modeling dataset is additionally restricted to one encounter per patient, as implemented in the Phase 2 preprocessing notebook."* That step remains necessary and should stay in the plan, because it prevents information leakage across the split. It does not, however, address the correlation among repeated observations within a partition. A model trained on 60,000 correlated rows from 45,000 patients does not have the statistical power of 60,000 independent observations, and `encounter_id`, which serves only as a chronology proxy, has no mechanism to correct for this. Preventing leakage and correcting for clustering are two separate problems, and only the first is currently addressed in the proposal and EDA report.

Path taken, and the result

There is direct precedent for resolving this in the source literature. Strack et al. (2014), the paper that produced this dataset, faced the identical issue and restricted their published analysis to one encounter per patient specifically to preserve independence between observations. A more recent methods paper, Zhao, Tanner, Golden, Fisher, and Rubin (2020), is written directly on this problem and demonstrates that common sampling and modeling approaches which ignore clustering in readmission data produce misleading results (see the added Works Cited entry at the end of this addendum).

Consistent with that precedent, the Phase 2 preprocessing notebook (Step 10, "Construct the primary patient-level analysis dataset") restricts the modeling dataset to one encounter per patient: each patient's first eligible encounter after the in-hospital-death exclusion in Step 5. Every backward-looking utilization field (`number_outpatient`, `number_emergency`, `number_inpatient`) still reflects the year before

that snapshot encounter, so no predictive history is lost by this restriction; only the correlated, repeated-encounter rows are removed.

Metric	Before (encounter-level)	After (patient-level snapshot)
Rows in modeling dataset	100,114 (post death-exclusion)	70,439
Reduction	—	29.6% of rows removed (repeat encounters, not lost patients)
30-day readmission rate	11.34%	8.94%

Table 3.5b. Effect of restricting to one encounter per patient, as run in the Phase 2 preprocessing notebook, Step 10.

The drop from 11.34% to 8.94% confirms the concern raised above: repeat patients were disproportionately responsible for the originally reported positive class, so the dataset's true, independent-observation base rate is lower than Figure 1 in this report shows. Any later phase that cites a class-imbalance figure should cite 8.94%, not 11.2% or 11.34%, since those are computed on correlated encounter-level rows.

Two further decisions were made alongside this change, both documented with reasoning in the same notebook:

- The previously planned `days_since_prior_encounter` engineered feature was dropped (Step 10 revision note), because the `encounter_id` ordering it would have been built from does not reliably measure elapsed calendar time, and the retained utilization fields already carry more direct signal. This is a correction to the data dictionary in both the proposal and this report, addressed in Section D below.
- A second, full-longitudinal, patient-grouped split was preserved as an explicitly labeled stretch-goal benchmark only (Step 14), not used to train the production model, so a later phase can quantify how much predictive performance and feature-importance stability change when repeated encounters are treated as independent rows versus collapsed to one per patient.

The split was validated with an explicit assertion confirming zero patient overlap between the training and test partitions (Step 12 output), and the resulting positive-class rate is close between partitions (9.64% train, 6.82%

test), though the gap is wider than in the original encounter-level split and is worth a monitoring note in the next model-evaluation writeup rather than a fix at this stage.

4. Discussion & Next Steps

4.1 Key Takeaways

Three findings from this exploratory pass are the most consequential for the project going forward. First, the severe class imbalance documented in Figure 1 (11.2% positive class) confirms that the evaluation plan laid out in the proposal — leaning on AUC-ROC and recall rather than accuracy — is the right call, and reinforces that class-imbalance handling (class weighting, and potentially resampling as a comparison) will need to be a first-class step in preprocessing, not an afterthought. Second, the data lends early, genuine support to the project's hypothesis: readmitted patients show higher utilization on every variable examined in Table 3, and the relationship between prior inpatient visits and readmission rate in Figure 5 is the strongest single pattern in the entire analysis, rising more than four-fold from patients with no prior admissions to patients with five or more. This is an encouraging signal to carry into feature engineering and modeling. Third, the collinearity check in Figure 6 shows no continuous predictor pair correlated highly enough to force an early feature-dropping decision, which keeps the feature set flexible heading into modeling, though the modest weak-linear/stronger-non-linear pattern in `number_inpatient` (Pearson $r = 0.17$ vs. the step-function shape in Figure 5) is a useful early argument for prioritizing tree-based models over a purely linear baseline. A fourth takeaway comes from the categorical side of the data (Tables 2 and 4): demographic predictors (gender, race) separate readmission risk only weakly, while treatment-intensity predictors — whether a diabetes medication was prescribed at all, and whether the regimen was changed mid-encounter — separate it more clearly, in the same direction as the utilization hypothesis. This

suggests the eventual feature set should lean on treatment and utilization signals rather than demographics as the primary risk drivers, which is also consistent with how the model is intended to be used: care teams act on modifiable clinical signals, not on demographic categories. Finally, the missingness audit in Tables 5 and 6 confirms that missingness in this dataset is not random or evenly spread — it clusters heavily in a handful of lab-result and specialty fields, and several coded fields carry a second, hidden layer of "unknown" values that a naive null count would miss entirely. Because several of the most-missing fields (`max_glu_serum`, `A1Cresult`) look more like a testing-protocol signal than noise, the cleaning plan below treats them as informative missingness to be encoded, not simply imputed or dropped.

None of these findings changes the underlying research question or hypothesis stated in Section 1. If anything, the utilization-based hypothesis looks more directly supported by this first look at the data than anticipated, particularly for prior inpatient visits. The one open question this analysis raises rather than answers is the unexpected readmission-rate spike in the youngest adult bracket (Figure 4); because that bracket is small, it will be revisited once a proper train/test split is in place rather than acted on now.

4.2 Data Cleaning & Preprocessing Plan

This EDA surfaced several concrete issues that revise and sharpen the cleaning plan described in the original proposal. None of these have been acted on yet — cleaning is intentionally deferred to the next phase — but each is now a scoped, specific task rather than a general intention.

- Structural leakage from in-hospital deaths: 1,652 encounters (1.62% of the dataset) end in patient death (discharge disposition codes 11, 19, 20, and 21). Every one of these rows is

coded "not readmitted" by construction, since a deceased patient cannot be readmitted — meaning these rows are guaranteed negative-class examples for a reason that has nothing to do with clinical risk. Consistent with the original dataset publication (Strack et al., 2014), these encounters will be excluded during cleaning rather than left in as ordinary negative examples, to avoid quietly biasing the model toward an artificially easy negative class.

- Non-independence of repeated encounters: 46.2% of encounters (from 23.5% of patients) are repeated observations from the same patient, and these repeat-patient encounters carry a 30-day readmission rate roughly five times that of single-encounter patients (19.55% vs. 3.96%). Consistent with the original dataset publication's own methodology (Strack et al., 2014), the modeling dataset has been restricted to one encounter per patient, using each patient's first eligible encounter, to preserve independence between observations. This was implemented in the Phase 2 preprocessing notebook (Step 10) and reduced the usable dataset to 70,439 rows, shifting the positive-class rate from 11.34% (encounter-level) to 8.94% (patient-level). A full-longitudinal, patient-grouped split was retained separately as a stretch-goal benchmark only.
- High and hidden missingness: Standard missingness (the "?" placeholder) ranges from negligible for `diag_1` (0.02%) up to 96.86% for `weight`, 94.75% for `max_glu_serum`, and 83.28% for `A1Cresult`, with `medical_specialty` (49.08%) and `payer_code` (39.56%) also substantial. `Weight` has already been dropped as non-informative at this sample size. The remaining high-missingness fields (`max_glu_serum`, `A1Cresult`, `medical_specialty`, `payer_code`) will need a deliberate strategy — most likely a "tested/not tested" or "specialty known/unknown" indicator rather than imputing a specific lab value or specialty that was never recorded. Separately, three coded fields (`admission_type`,

discharge_disposition, admission_source) contain hidden missingness in the form of literal category labels such as "NULL," "Not Available," and "Not Mapped," affecting 10.22%, 4.60%, and 6.66% of encounters respectively. These will not surface in a standard null-count and must be handled explicitly as their own missing/unknown category rather than treated as informative labels.

- High-cardinality diagnosis codes: diag_1, diag_2, and diag_3 are ICD-9 codes with hundreds of distinct raw values. As planned in the proposal, these will be grouped into a smaller number of clinically meaningful categories before encoding rather than one-hot encoded directly, to avoid a feature space that is large relative to the number of available rows.
- Class imbalance: With the positive class at roughly 11%, preprocessing will apply class weighting as the primary approach, with resampling methods such as SMOTE evaluated as a comparison rather than a default, consistent with the proposal.
- Encoding and leakage guardrails: Remaining categorical fields (admission type, discharge disposition, admission source, age bracket) will be encoded for modeling, and any field only known after discharge will be explicitly dropped or masked so it cannot leak post-outcome information into the predictors.
- Time-aware, patient-grouped validation: Because the public release does not include exact calendar dates or hospital identifiers, encounter_id will continue to serve as an ordinal proxy for chronological order, with the earliest ~75–80% of encounters forming the training set and the most recent ~20–25% held out as a test set. Splits will additionally be grouped by patient_nbr so that no single patient's encounters appear in both partitions. This design choice is unchanged by the EDA and remains necessary

given the dataset's structural limitations.

Taken together, these steps update the preprocessing plan from a general list of intentions into a checklist with specific columns, specific row counts, and specific justifications, which should make the next phase of this project — feature engineering and model training — considerably faster to execute correctly the first time.

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Appendix

A. Data Dictionary and Codebook

The table below lists every variable that exists in the model-ready table produced by this phase, following the same format as the data dictionary in the EDA Report's appendix. Engineered variables are marked as such; they do not exist in the raw UCI download and were constructed during preprocessing or feature engineering as described in Sections 2 and 3.

Variable	Data Type	Source / Status	Description
patient_nbr	Integer (ID)	Source	Unique patient identifier; used as the deduplication and grouping key for the patient-level snapshot.
encounter_id	Integer (ID)	Source	Unique identifier for a single inpatient encounter; underlies the ordinal chronology proxy.
encounter_order	Integer	Engineered	Sequential rank derived from encounter_id, used as a stand-in for calendar time in the split.
readmitted_30d	Binary (0/1)	Engineered / TARGET	Collapsed from the native three-class label; 1 if readmitted within 30 days.
age	Categorical	Source	Ten-year age bracket.
race	Categorical	Standardized	Includes an explicit missing category for unrecorded values.
gender	Categorical	Source	Patient gender classification.
admission_type_id_label	Categorical	Mapped / clean	Descriptive admission type; hidden-unknown placeholders consolidated.
discharge_disposition_id_label	Categorical	Mapped / clean	Descriptive discharge destination; hidden-unknown placeholders consolidated.
admission_source_id_label	Categorical	Mapped / clean	Descriptive referral source; hidden-unknown placeholders consolidated.
time_in_hospital	Integer	Source	Length of stay in days.
num_lab_procedures	Integer	Source	Number of lab tests performed during the encounter.
num_procedures	Integer	Source	Number of non-lab procedures performed during the encounter.
num_medications	Integer	Source	Number of distinct medications administered during the encounter.
number_outpatient	Integer	Source	Outpatient visits in the year prior to the encounter (already lagged).
number_emergency	Integer	Source	Emergency visits in the year prior to the encounter (already lagged).
number_inpatient	Integer	Source	Inpatient admissions in the year prior to the encounter (already lagged).
diag_1_group / diag_2_group / diag_3_group	Categorical	Engineered	Primary, secondary, and tertiary diagnoses grouped into 9 clinical categories.
number_diagnoses	Integer	Source	Total number of diagnoses recorded for the encounter.
max_glu_serum_tested	Binary categorical	Engineered	Whether a glucose serum test was ordered (Tested / Not Tested).
A1Cresult_tested	Binary categorical	Engineered	Whether an HbA1c test was ordered (Tested / Not Tested).

Variable	Data Type	Source / Status	Description
patient_nbr	Integer (ID)	Source	Unique patient identifier; used as the deduplication and grouping key for the patient-level snapshot.
encounter_id	Integer (ID)	Source	Unique identifier for a single inpatient encounter; underlies the ordinal chronology proxy.
change	Categorical	Source	Whether diabetic medication dosage was altered during the visit.
diabetesMed	Categorical	Source	Whether any diabetic medication was prescribed.

Table A1. Data dictionary for the model-ready table produced by this phase.